Bioinformatics

2025/2026

Day 4

Exercise 1

Use conda to create an environment named bioinfo_lab4 that includes the latest version of samtools and the necessary software for all exercises. Create an environment.yml file with the versions used.

Exercise 2

Downloading entire Whole Genome Sequencing (WGS) BAM files is computationally inefficient for localised analysis. You will use **samtools** to query a specific genomic coordinate directly from the 1000 Genomes Project server using the BAM index (**.bai**).

2.1 Open your terminal and use samtools to extract reads from chromosome 20, positions 1,000,000 to 1,000,500, from a Phase 3 sample (HG00096) that may be found in https://ftp.1000genomes.ebi.ac.uk/vol1/ftp/phase3/data/HG00096/alignment/HG00096.mapped.ILLUMINA.bwa.GBR.low_coverage.20120522.bam. Ensure your command retains the original SAM headers in the output. Save the resulting file locally as ***subset_region.sam***.

2.2 Inspect the header section of your generated subset_region.sam file. Locate the appropriate Program (@PG) record to identify the specific alignment software and the exact version used to generate the original BAM file.

2.3 Locate and list the filenames of any precomputed variant files (specifically Indels or Genotypes in .vcf.gz format) associated with this specific sample. Hint: examine the header of the downloaded SAM file.

2.4 The second column of a SAM file contains a decimal integer representing a bitwise flag. You must use base-2 logic to isolate specific read populations.

2.4.1 Calculate: What is the integer value for a read that meets the following criteria?

- Read is paired
- Read is mapped in a proper pair
- Query read maps to the reverse strand

2.4.2 Use samtools view with the -f (include) and/or -F (exclude) parameters to extract only the reads matching the exact criteria from subset_region.sam using the criteria from Question 2.4.1. Save the output to reverse_proper_pairs.sam. How many reads do you get?

2.4.3 Create a file **mate_unmapped.bam** with all reads where the *mate is unmapped*, regardless of other flags.

2.5 Search your subset_region.sam file for the alignment record with the longest CIGAR string (defined by the total number of characters in the string) that includes at least one insertion (I) or deletion (D).

Tie-breaking Rule: If multiple records share the same maximum character length, select the string that is lexicographically smallest.

Record the full CIGAR string and calculate the total number of "matches/mismatches" (M) versus "indels" (I or D) for this specific read.

1. **Read Length:** Calculate the exact length of the sequencing read (the query sequence) that produced this CIGAR string. Show your arithmetic.
2. **Reference Span:** Calculate the exact number of base pairs this read spans on the reference genome. Show your arithmetic.

2.6 Format Conversion and Indexing SAM files are human-readable but computationally inefficient for large-scale queries. You must convert your subset to a compressed binary format (BAM) and generate an index to enable coordinate-based visualisation.

2.6.1 Convert your subset_region.sam to a BAM file named subset.bam.

2.6.2 Sort the alignments by genomic coordinates (a requirement for indexing).

2.6.3 Generate the index file (subset.bam.bai).

2.7 A "pileup" summarises the base calls from all reads covering a specific genomic position. This allows for the identification of variants (SNPs/Indels) relative to the reference.

2.7.1 Use `samtools mpileup` and `samtools tview` to generate a pileup for the region you extracted. Focus the analysis on chromosome 20, position 1000000 up to 1000100.

2.7.2 Can you locate a position where multiple reads show a base different from the reference? Can you identify a high-confidence SNP or Indel in this 100-bp window? Describe how it is represented in the pileup output (hint: you may need to download the reference genome used and index it with *samtools faidx)*.

Exercise 3

In the previous exercises, you manipulated pre-aligned data. You will now perform the alignment process yourself using minimap2 and bowtie2 aligners.

3.1 Data Preparation.

3.1.1 Download a small genome as reference.fa.

Location:
https://raw.githubusercontent.com/BenLangmead/bowtie2/master/example/reference/lambda_virus.fa

3.1.2 Download a small set of FASTQ reads as ***reads.fq***
Location:
https://raw.githubusercontent.com/BenLangmead/bowtie2/master/example/reads/reads_1.fq

3.2 Align the FASTQ reads against the reference using minimap2 and output directly to a sorted BAM file: minimap2_alignment.bam file (hint: use a pipe to samtools sort). How many reads align?

3.3 Align the FASTQ reads against the reference using bowtie2 and output directly to a sorted BAM file: bowtie2_alignment.bam file. How many reads align? Hint: you will need to index the reference before aligning.

3.4 How does the number of secondary alignments compare between minimap2 and bowtie2? You should observe a count of 0 for both.

Exercise 4

You got an insertion of 5 consecutive adenines (AAAAA) in your sample relative to the reference. You are configuring an alignment algorithm to score this event.

Model A (Linear Gap Penalty): Uniform cost of **2** per gap position.

Model B (Affine Gap Penalty): Uses the function **W(k) =** γ **+ δ □ k**, where gap opening penalty γ **= 8**, and gap extension penalty **δ = 0.5**.

4.1 Calculate the total penalty cost for this 5-bp insertion under Model A.

4.2 Calculate the total penalty cost for this 5-bp insertion under Model B.

4.3 Why is the affine model biologically more realistic than the linear model for a 5-bp insertion?

Exercise 5

You have isolated a mystery sequence from an environmental sample.

Query Sequence: `ATGAGCAAGAAAATTGTCATCGTCGGCGCCGGCGTGGCCGGCAGCACCGCCGCT`

- Navigate to NCBI blastn.
- Paste the sequence and run the search against the **Standard Databases (nr/nt)**.

5.1 What is the E-value of the top hit?

5.2 The top hit likely shows 100% identity. Does high identity *guarantee* homology, or is it an inference? Explain.

Exercise 6

The web browser is rarely used for large-scale analysis. Instead, we utilise the **BLAST+ command-line suite**. This exercise simulates the setup of a localised alignment server.

6.1 Environment Setup and Installation

First, ensure the BLAST+ binaries are available in your specific laboratory environment.

6.1.1 Use conda to install the BLAST+ suite into your current environment:
6.1.2 Confirm the installation by checking the version of the nucleotide-to-nucleotide aligner (blastn).

6.2 Data Acquisition: Reference and Query. To run a local BLAST, you need a subject (the database) and a query (the sequence to search).

Reference Database: Download the complete genome of *Escherichia coli* (or use the reference.fa from Exercise 3).

Query Sequences: Create a file named query.fasta containing the following sequences:

```
>mystery_read_1
GGTAAAGCGGCAAAAAACTACCGTGAAAAGTCGGTGGATGTGGCGGGTTATGATGAACT
>mystery_read_2
GTCTCCTCTGACTTCAACAGCGACACCCACTCCTCCACCTTTGACGCTGGGGCTGGCA
```

6.3 Raw FASTA files are not searchable by BLAST. You must index them into a specialised binary format using makeblastdb tool.

After running the command, list your files (ls). You should see files ending in .nhr, .nin, and .nsq. What is the specific purpose of these three file extensions in the BLAST+ index?

6.4 With the database prepared, you can now execute the search locally without an internet connection.

Run the Search

- blastn -query query.fasta -db ecoli_db -out blast_results.txt -outfmt 6
- *Note: -outfmt 6 produces a tab-delimited table, which is the standard for automated parsing.*

Inspect the blast_results.txt file and identify the **E-value**. What are the top 5 hits with lowest E-Value?

6.5 Building a database like nr (non-redundant proteins) or nt (nucleotides) from a raw FASTA file can take days and massive RAM. Instead, NCBI provides pre-formatted, indexed versions of these databases for direct download.

6.5.1 Locate the **update_blastdb.pl** script included in the BLAST+ suite (tip: use the which command).

6.5.2 Download and decompress the "RefSeq Select" RNA database (requires about 250 MB of disk space). List the file extensions created after decompression. How do these differ from the files created when you ran makeblastdb manually?

6.5.3 Perform a local search against this newly acquired database using the query.fasta created previously. Is the top hit the same as your previous search in Exercise 5.2?

6.5.4 The ***nt*** database currently exceeds 100 GB when compressed. Why would a research institute choose to store this locally and update it weekly rather than simply using the NCBI web-interface for all their scientists?

Consider: Search speed, data privacy, and the ability to integrate BLAST into automated pipelines.